

# **Information Technologies for Industrial Engineers**

# JavaScript Frameworks

# What is a framework?

- A framework is a library that offers **opinions** about how software gets built.
- These opinions allow for **predictability** and **homogeneity** in an application.
  - Predictability allows software to scale to an enormous size and still be maintainable.
  - Predictability and maintainability are essential for the health and longevity of software.

# Imperative and declarative programming

- **Imperative**
  - Tell a computer how to do something.
- **Declarative**
  - Describe to a computer what you want.

# To do this

- List 1
- List 2
- List 3

# Imperative programming

- JavaScript

```
const ul = document.createElement("ul");
for (let i = 1; i < 4; i++) {
  const li = document.createElement("li");
  li.textContent = `List ${i}`;
  ul.appendChild(li);
}
document.body.appendChild(ul);
```

# Declarative

- Just use HTML

```
<ul>  
  <li>List 1</li>  
  <li>List 2</li>  
  <li>List 3</li>  
</ul>
```

## But...

- `HTML` only works for static site.
- For dynamic web application, a **framework** can help us write code more **declaratively**.



# Vanilla JS or Framework?

- Vanilla JS
  - A less complex site, for example a personal to-do list or a site that displays mostly static content
- Framework
  - A large site with a complex UI. Frameworks provide solutions to common problems that would take an absurd amount of time and patience to implement with pure JavaScript.

# Framework popularity

- State of JS 2025
- Stack Overflow Survey 2025

# Vue

# Vue Template Syntax

```
<template>  
  <h1>Hello, world!</h1>  
</template>
```

- Vue uses template syntax to describe UI declaratively.

# Vue Single-File Component (SFC)

- A Vue SFC is a file that encapsulates the template, logic, and styles of a component in a single file.

```
<script setup lang="ts">
import { ref } from "vue";

const count = ref(0);
</script>

<template>
  <button @click="count++">Count: {{ count }}</button>
</template>
```

# Comparison with Vanilla JS

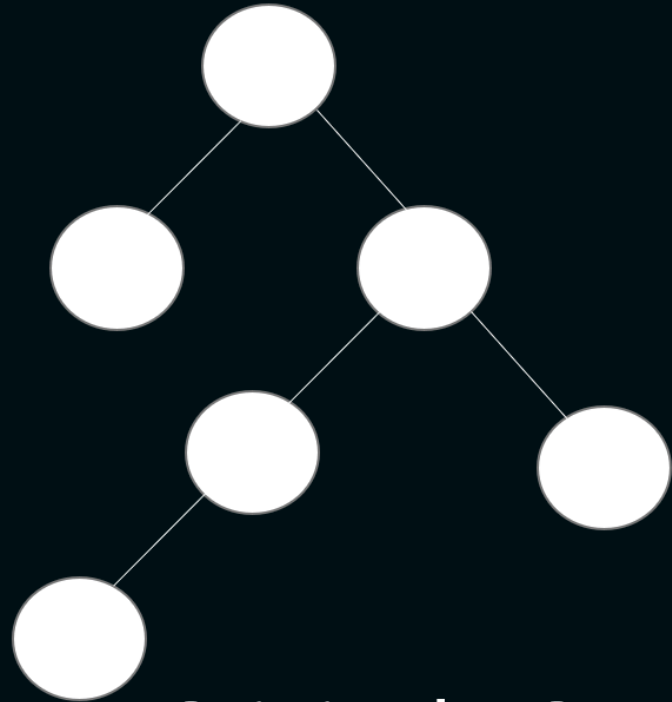
```
<button>Count: 0</button>
```

```
const btn = document.querySelector("button");  
let count = 0;  
function updateButton() {  
  btn.textContent = `Count: ${count}`;  
}  
btn.addEventListener("click", () => {  
  count++;  
  updateButton();  
});
```

[Link to CodePen](#)

# Virtual DOM

- Virtual DOM (VDOM) represents the DOM tree in memory.
- VDOM performs calculations and decides which elements of the actual DOM should be mutated.
  - Faster performance.
- [Source](#)



Original DOM



# Rendering process

- Vue uses a **reactive system** to track changes in the state of your application.
- When a reactive state changes, Vue automatically re-renders the affected components.
- Vue tracks dependencies and updates only the affected parts of the DOM.
- See illustration in [Source](#).

# Developing with Vue

- You need **Node.js** and a **package manager** (npm or pnpm) installed.
- Node.js is a JavaScript runtime that allows you to run JavaScript code outside of a web browser.
- A package manager is a tool that allows you to install and manage libraries and tools for your project.

# Install `node`

- We will use `nvm` (Node Version Manager) to install Node.js.
- Windows
  - Open terminal as administrator and run the following command:
  - `choco install nvm`
- MacOS
  - Follow the instructions to add `nvm` to your shell.
  - *(See next page)*

# Adding `nvm` to your shell (MacOS)

## 1. Create the NVM Directory

```
mkdir ~/.nvm
```

## 2. Configure Your Shell Profile

```
nano ~/.zshrc
```

Add the following lines to the end of the file:

```
export NVM_DIR="$HOME/.nvm"  
[ -s "$(brew --prefix nvm)/nvm.sh" ] && \. "$(brew --prefix nvm)/nvm.sh"  
[ -s "$(brew --prefix nvm)/etc/bash_completion.d/nvm" ] && \. "$(brew --prefix nvm)/etc/bash_completion.d/nvm"
```

## 3. Apply the changes `source ~/.zshrc`

# Install node

- All
  - `nvm install 24`
  - `nvm use 24`
  - `node -v` (check version)
  - `npm -v` (check version)

# Install `pnpm`

- We can use `npm`, but `pnpm` is a faster and more efficient package manager.
- `npm install -g pnpm`
- `pnpm -v` (check version)

# Test `node`

- Create file `test.js` with the following content:

```
console.log("Hello, world!");
```

- Run the file using `node test.js`

# Get started with `vue`

- `pnpm create vite@latest`
  - Project name : `my-app` (any name)
  - Select a framework : `Vue`
  - Select a variant : `TypeScript`
- `cd my-app`
- `code . -r` (*Set VSCode workspace*)
- `pnpm install`
- `pnpm run dev`



# Build

- `pnpm run build`

# Preview

- `pnpm run preview`

# Deploy

- Deploy `./dist` folder to Netlify.

# Online IDE

- <https://codesandbox.io/>
- <https://stackblitz.com/>